Serving the full model. Meeting the latency gate.

Qwen3-235B-A22B
PD disaggregated inference

VNUHCM - TripleU

Evidence snapshot | 12 October 2026

BEST INTERNAL RESULT

1,912.30

good output tokens/s
97.44% SLA fraction
N48 / 16 H200

Local measurements; organiser re-run and accuracy validation pending.

Best measured stack, with explicit limits

GOOD OUTPUT RATE

1,912.30

req/s goodput × 200
9.56150 good req/s

REQUESTS WITHIN BOTH SLAs

97.44%

1,180 / 1,211 requests
0 errors; 0 cancellations

VALIDATION STATUS

Pending

Full-set accuracy
Organiser official re-run

Evidence: job 22415 / confirm / N48; canonical AIPerf export.

Throughput counts only when latency passes

WORKLOAD

4,000 → 200

Synthetic Shakespeare inputs
Official FP8 checkpoint

PER-REQUEST SLA

3,000 / 20

TTFT ≤ 3,000 ms
Mean ITL ≤ 20 ms

POINT ACCEPTANCE

≥ 90%

Requests must pass both
Errors stay in denominator

Closed-loop load: a completed request is immediately replaced, sustaining N concurrent requests.

Task Rules + 2026-09-25 addendum, included under AI_Task/logs/rules/.

One platform, one unchanged checkpoint

COMPUTE	2 × HGX H200 nodes / 16 GPUs total
NETWORK	InfiniBand NDR between the two nodes
MODEL	Qwen3-235B-A22B-Instruct-2507-FP8
SERVING	SGLang 0.5.17 / router 0.3.2 / NIXL 1.3.2
CLIENT	AIPerf 0.12.0; pinned official tokenizer
PRECISION	Published FP8 weights / BF16 KV (auto, verified in logs)

Image metadata, SHA256 and model metadata: AI_Task/job_scripts/environment.json.

Best layout: 2 prefill + 1 decode worker

NODE 0 / 8 H200

Prefill worker 0 · 4 GPUs · TP4 / EP1

Prefill worker 1 · 4 GPUs · TP4 / EP1

KV cache NIXL →

NODE 1 / 8 H200

Decode worker 0
8 GPUs · TP8 / EP8

AIPerf N48 → OpenAI-compatible gateway :8000 → Prefill → Decode → streamed tokens

P8-TP4_D8-TP8-EP8 / N48; hetero-tp-request-descriptors-v1; job 22415.

Starting layouts establish our measured reference

Reference layout 07

Prefill: 4 workers × TP2
Decode: 2 workers × TP4 / EP4
Node split: 8 + 8 GPUs
Long run: N32

Previous best layout 09

Prefill: 2 workers × TP4
Decode: 2 workers × TP4 / EP4
Node split: 8 + 8 GPUs
Long run: N40

Trade-off: fewer prefill replicas, wider tensor parallelism; measure the balance under the SLA.

07 and 09 are organiser-provided starting layouts; the team extends the search and transport.

A 17-layout search, with measured coverage

GPU SPLIT P + D	PREFILL TP	DECODE TP	LAYOUTS
8 + 8	2 / 4 / 8	2 / 4 / 8	9
4 + 12	2 / 4	2 / 4	4
12 + 4	2 / 4	2 / 4	4

14/17 originals measured

+ 3 measured EP2 variants

Each completed topology has N16, N32, N40 and N48 actual measurement exports.

AI_Task/logs/COVERAGE.md; pending and failed runs are not counted as completed measurements.

Screen quickly; confirm at the full duration

Deploy + initialize

All P/D paths
Separate setup prompts

Explore

N16 / 32 / 40 / 48
10s + 20s + 30s

Long confirmation

Best N48 / latest N32
60s + 120s + 30s

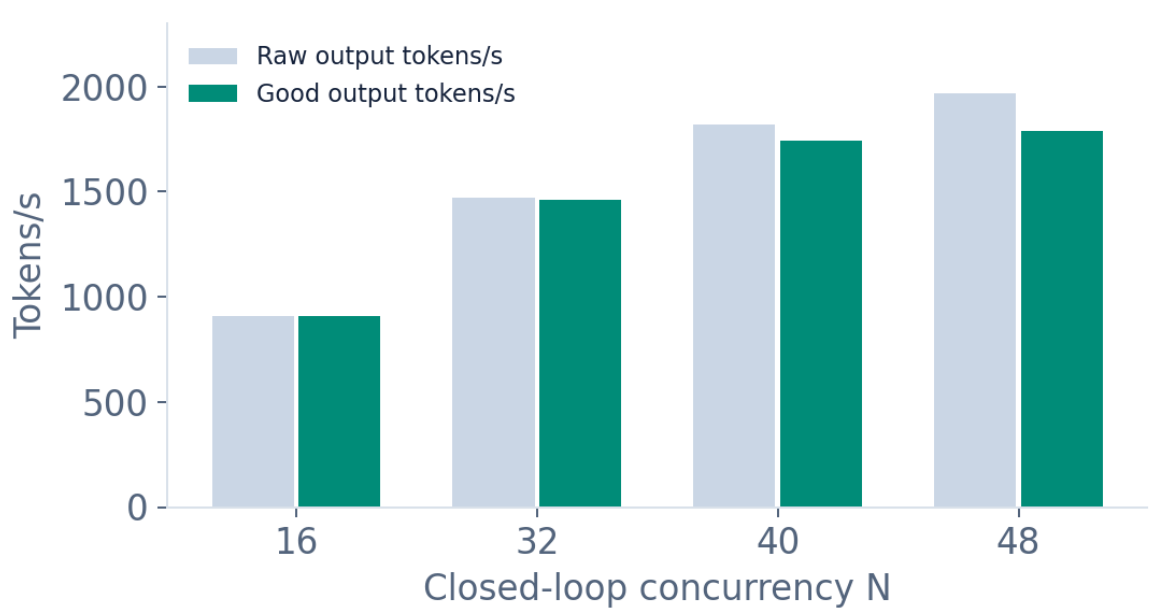
Official timing

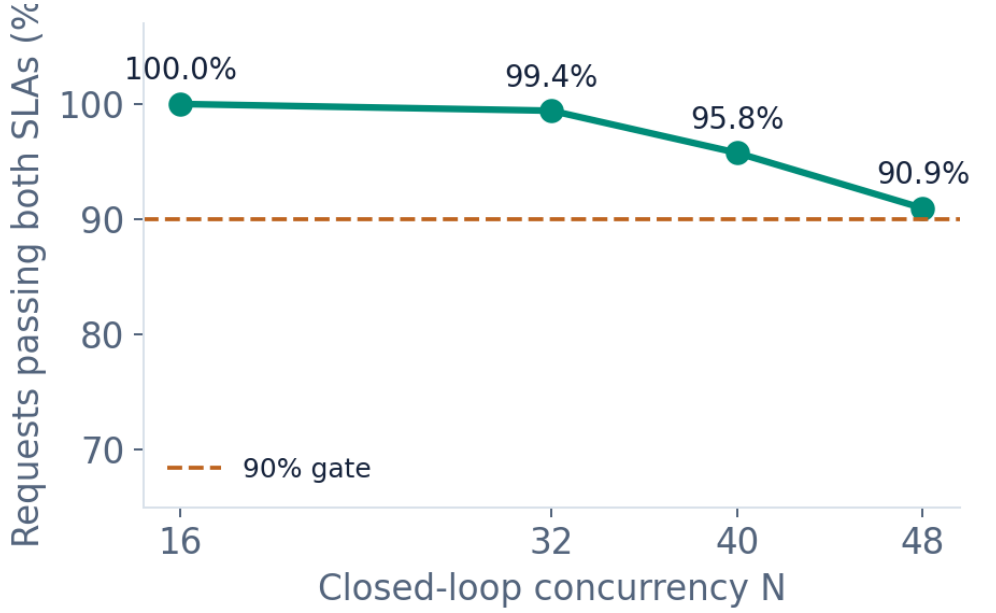


A short sweep and a long confirmation use different windows; their difference is not an optimisation gain.

Exploratory 10/20/30 vs long 60/120/30; original canonical exports retained.

Winner sweeps select N48; the long run confirms it

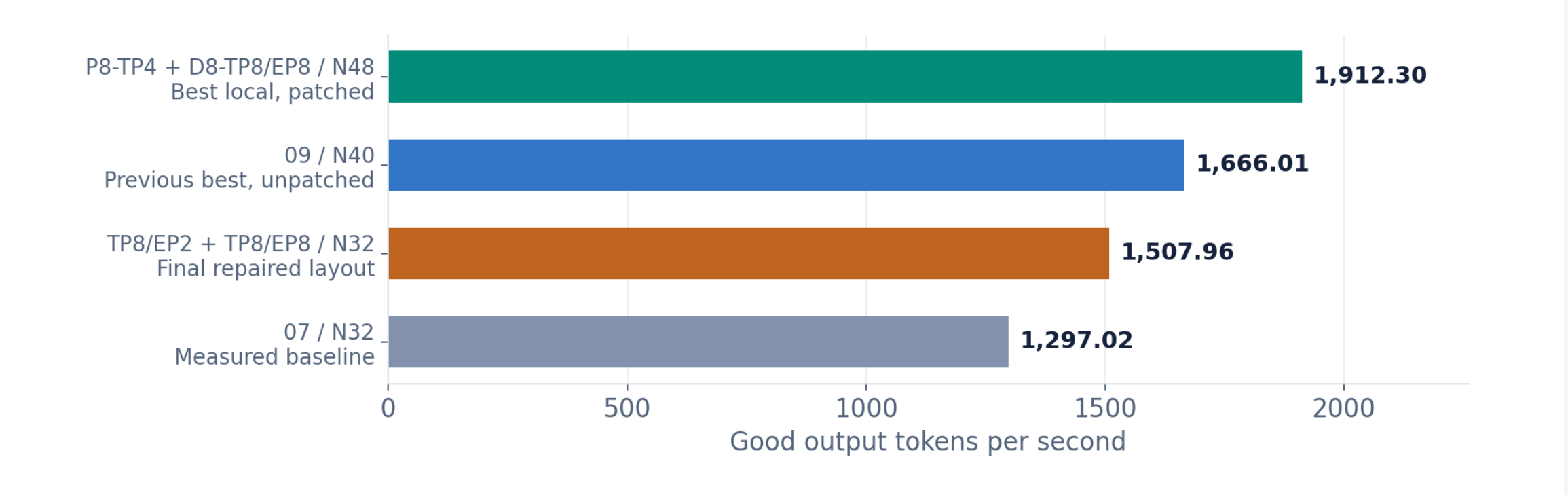




N48 sweep: 1,788.65 good tokens/s; 90.95% SLA. Long N48: 1,912.30; 97.44% SLA.

Job 22415 exploratory sweep: 10s warmup / 20s measurement / 30s grace.

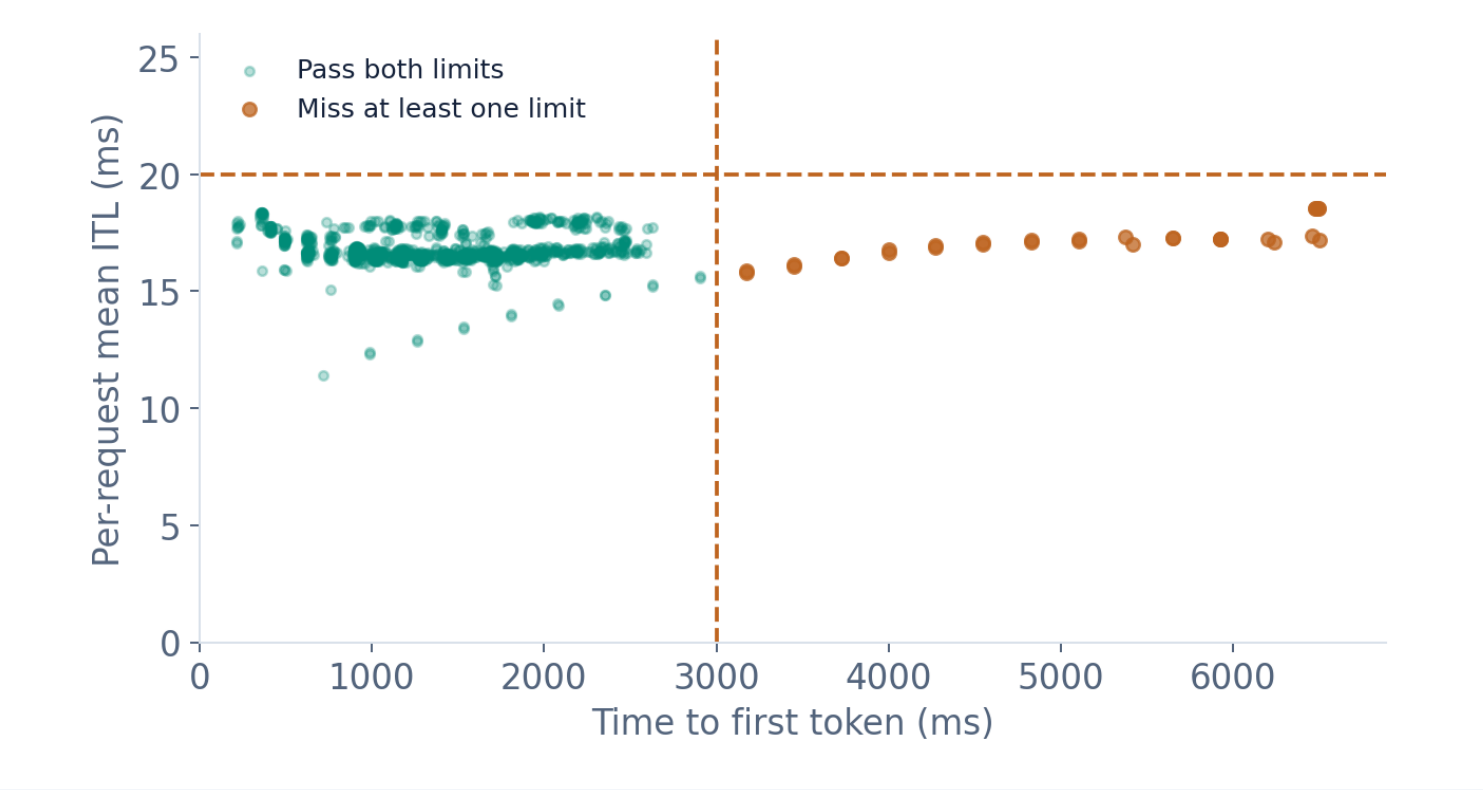
Best long run: +14.78% over the previous best



Observed gain includes topology, concurrency and transport changes. Repeated matched A/B is still needed.

Jobs 22415, 22219, 22809 and 22197: long 60/120/30 runs on 16 H200.

Check the gate per request, not from averages



1,180 / 1,211

requests pass both limits

Mean TTFT: 1,482.22 ms
Mean ITL: 16.79 ms
P95 TTFT: 2,456.56 ms
P95 ITL: 17.99 ms

Job 22415 / confirm N48: 1,211 profiling-phase records in the original JSONL.

Failures remain evidence; actual exports count

FP8 sharding	TP8 / EP1 gives a 192-wide expert shard; FP8 block size is 128.	Use a distinct EP2 variant; keep the EP1 original incompatible.
Cold P/D transfer	22778 timed out before measurement; patched 22809 completes.	Request-scoped descriptors; setup deadlines do not relax benchmark SLAs.
Distributed ports	A deployment hit EADDRINUSE before producing metrics.	Assign explicit distinct ports; preserve the failed logs and source.

Immutable payloads and failed logs remain included; recovery registry distinguishes EP2 variants.

EP2 makes TP8 compatible with FP8 blocks

TP8 / EP1

128 experts / 1 expert group
1536 ÷ 8 = 192
192 is not divisible by 128

TP8 / EP2

64 experts / group × 2 groups
1536 ÷ 4 = 384
384 = 3 × 128

All 3 EP2 replacements are measured. All experts and weights are retained; the originals stay labelled.

Checkpoint: MoE intermediate size 1536, 128 experts; image loader requires block_n = 128.

NIXL: two request-scoped transport patches

Full-pool descriptor plan

≈ 403.8 million source entries / rank

Request-scoped plan

188 entries: contiguous TP8→TP4 or TP8→TP8

Byte ranges, head mapping and notifications retained.

720 + 936

CPU range cases
Both patches GPU-tested

1,507.96

latest equal-TP / N32
95.38% SLA; no errors

Heterogeneous TP: jobs 22294 / 22415. Equal TP8: job 22809. Original source and diffs included.

Measured winner and portable replay are distinct

01	**Current winner**	Job 22415 / N48; source, patch, worker logs and canonical exports.
02	**Portable entry**	submit.sbatch still replays unpatched 09 / N40, job-22219 layout.
03	**Frozen audit trail**	All historical attempts and source hashes; raw measurements unchanged.
04	**Next replay step**	Port the winning patch/layout and GPU-validate the packaged entry.

AI_Task/job_scripts/submit.sbatch; archive/22415; supplementary/; BUILD_AND_RUN.md.

Correctness claims follow the available evidence

Published FP8 weights and BF16 KV, unchanged	**Worker logs / source**
Plain PD on 16 GPUs; no speculative decoding	**Configured / local logs**
Best long run / N48 latency gate	**97.44% pass fraction**
Full-set accuracy ≥ 99% of reference	**Not yet tested**
Organiser reconstruction + official scored re-run	**Pending**

Local latency evidence is available. Official acceptance and any Bonus validation remain pending.

Qwen addendum Q1/Q2; accuracy pass line: at least 99% of organiser reference score.

Topology coverage complete; submission work remains

Measured coverage

14 original + 3 EP2 layouts
All four concurrency sweeps
Winning long-run evidence
Both NIXL patches and logs

Still outstanding

Full-set accuracy check
Portable winning-stack replay
HPC/OpenFOAM evidence
Official upload and re-run

Snapshot: 2026-10-12T10:24:13.693598+00:00. Job 22809 completed 11 October, 23:18 UTC.

Current winner: TP4 prefill + TP8 / EP8 decode

1,912.30

good output tokens per second

97.44% request SLA fraction · N48 / 16 H200 · Official FP8 model

Accuracy and organiser re-verification remain pending.

Job 22415 / N48; hetero-tp-request-descriptors-v1; long 60/120/30 measurement.

Appendix: the final repaired layout and selection

Final layout	P8-TP8-EP2_D8-TP8-EP8; job 22809 completed.
N32 long run	1,507.96 good tokens/s; 95.38% SLA; no errors/cancels.
N40 / N48 short sweeps	89.50% / 86.06% SLA: both miss the 90% gate.
Selection	Job 22415 remains the best passing long measurement.
Interpretation	Measured coverage does not mean every layout passes SLA.

Job 22809 / equal-tp8-request-descriptors-v1; sweeps 10/20/30 and confirm 60/120/30.